

Draft - expert review pending

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Topic 3.3 - Hazards in health, safety and environment

A real classroom situation

After the 2022 floods, a school in Dadu became a relief camp. Within a week, the head teacher had dealt with a scorpion bite, a child's cut from a broken window, a gas cylinder moved next to the cooking fire, and two volunteers who could not sleep. When a health officer visited, she said, "You are running a small HSE operation, you know." The head teacher had never heard the term. "I call it Tuesday," she said.

This topic names what she was doing. Health, safety and environment, HSE, is the field that studies hazards to people and the environment together, and it begins with one word used precisely: hazard.

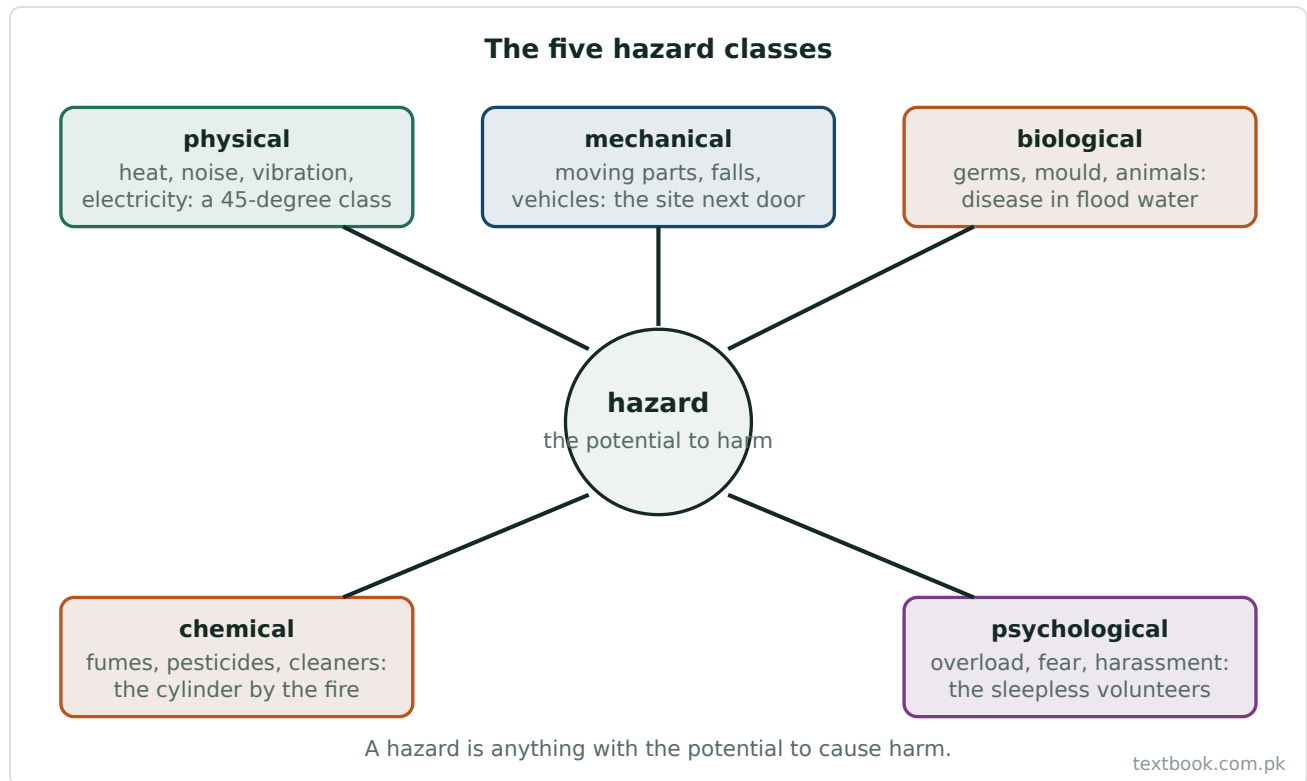
Explanation

Health, safety and environment: hazards

A hazard, in the HSE sense, is anything with the potential to cause harm. The word names the thing or condition, not the event: the wet floor is the hazard; the slip is the incident. Health, safety and environment as a field exists because the three belong together: the same factory smoke is a safety hazard to the worker at the furnace, a health hazard to the neighbourhood, and an environmental hazard to the crops downwind. The World Health Organization describes occupational health as the promotion of the highest degree of

physical, mental and social well-being of workers, and lists heat stress, airborne particulates and mental health among its concerns (WHO, n.d.).

Hazards sort into five classes, and the sorting matters because each class is controlled differently.



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- **Physical hazards** are energy and conditions: heat, noise, vibration, poor lighting, electricity. A Sindh summer classroom at 45 degrees is a physical hazard, and so is the generator exhaust beside it.
- **Mechanical hazards** come from moving things and falls: machinery parts, vehicles, ladders, collapsing structures. The construction site next door is full of them.
- **Biological hazards** are living things and their products: bacteria in flood water, mould after damp, insects, hepatitis in clinics.
- **Chemical hazards** are substances: cleaning acids, pesticides, fumes, dusts. A farm spraying season is a chemical-hazard season.
- **Psychological hazards** are conditions that harm the mind: unmanageable workload, fear, harassment, disaster stress. The two sleepless volunteers in Dadu had a real hazard exposure, by the same standard as the scorpion.

The five hazard classes, with controls

Class	One example	Health effect	One safety measure
Physical	a 45-degree classroom; generator exhaust beside it	heat exhaustion; carbon monoxide	shade, water, breaks; exhaust routed away
Mechanical	a collapsing wall; unguarded machinery	cuts, crush injuries, falls	barriers; guards; tagging out the danger
Biological	flood water; mould; insects and scorpions	water-borne disease; bites; infection	safe water; drying out; clearing debris
Chemical	cleaning acids; pesticide spray; a cylinder by a fire	burns, poisoning; fire and blast	labels; storage apart; directions followed
Psychological	displacement stress; unmanageable workload	sleeplessness, anxiety, breakdown	rotation, rest, someone to talk to

The quiet exposures, heat, dust, noise, stress, carry as much burden as the dramatic accidents.

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A Karachi study of a construction site found its most common hazards were excavation work, unsafe working practices and missing or unused protective equipment, alongside faulty, unmaintained machinery (Abbasi et al., 2022). Notice what that list is: one mechanical source, and then human and organisational factors. Real hazard profiles are always mixed.

Health and environment; environmental safety

Health and environment are one subject seen from two ends. The environment is where the exposures come from; health is where they arrive. The same dust that dirties a classroom carries PM2.5 into lungs; the same flood water that ruins a house carries disease into a relief camp. Environmental safety is the management of that connection: keeping the surroundings, the school, the worksite, the water, the air, in a state that does not injure the people inside them.

A worked example. Take the Dadu school, one week, all five classes: physical, heat and standing water; mechanical, the broken window and a collapsing wall section; biological, water-borne disease and insects; chemical, the cylinder by the fire and bleach stored with food; psychological, displacement and loss. The

head teacher's week, sorted, becomes a hazard register, which is Topic 3.4's tool.

Common misconception. "Hazards mean dramatic accidents." Falls and fires are real, but the burden runs through the quiet exposures: heat, dust, noise, stress, contaminated water. The five-class sorting exists precisely so the quiet ones are not missed because the loud ones get the attention.

Activity: Hazard-spotting walk

Work in pairs. With permission, walk one loop of the school building and its yard, inside and out. Find at least one hazard in each of the five classes, write each on a card with its class name, and rank your five cards from most to least likely to injure someone this term. Allow 25 minutes. Handle nothing; observe only.

Check your understanding

1. Define hazard, and distinguish it from an incident, with one example of each. (*Understand*)
2. Name the five hazard classes and give one example of each from a school. (*Remember*)
3. Why is "the two volunteers cannot sleep" a hazard finding and not a private matter? (*Analyze*)
4. The Karachi site study found hazards mostly in practices and missing protection, not machinery alone. What does that suggest about where controls belong? (*Analyze*)

Summary

HSE studies hazards to people and the environment together. A hazard is anything with the potential to cause harm, and hazards sort into five classes, physical, mechanical, biological, chemical and psychological, each controlled differently. Health and environment are one connection seen from two ends, and environmental safety is its management. Real hazard profiles mix all five

classes, and the quiet exposures, heat, dust, noise, stress, carry as much burden as the dramatic accidents.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can define hazard and distinguish it from an incident.
- ☐ I can name the five hazard classes with a school example of each.
- ☐ I can explain why psychological harms belong on a hazard register.
- ☐ I can read a real situation, like the Dadu week, and sort it into the five classes.

Try this at your practicum school

Do the hazard-spotting walk with your cooperating teacher's agreement. Bring back your five cards, one per class, and ask the caretaker which one they would fix first. Compare their answer with your ranking and note the difference.

Summative task

Write 170 to 200 words on the following.

Your school is used as a flood relief camp for a month, as the Dadu school was. Write a short hazard brief for the volunteers: name at least four hazards across at least three classes, and for each, one control a volunteer could apply the same day.

Mini-rubric

Criterion	What a strong response does
Classes used correctly (Apply)	Spans at least three classes, each named

Criterion	What a strong response does
Hazards realistic (Understand)	Chooses exposures a real camp would have
Controls same-day (Apply)	Each control is something a volunteer can do without budget
Clarity and register	Plain sentences for volunteers; within the word count

Further reading

- World Health Organization. *Occupational health* (health topic):
<https://www.who.int/health-topics/occupational-health>
- Abbasi, H. N., Khan, R., Nasir, I., & Ahmed, W. (2022). A case study of Tariq Road Underpass, Karachi: Environmental analysis. *International Journal of Economic and Environmental Geology*, 12(4).
<https://doi.org/10.46660/ijeeg.Vol12.Iss4.2021.651>
- Shilling, R. S. F. *Occupational Health Practice*. (guide's "Environmental Health Practice" closest real work, cited by reference only; flagged per D-2026-0001)

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